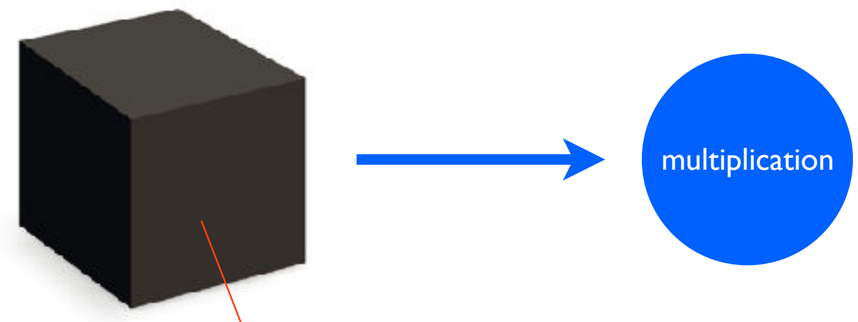
38. Cognitive Architecture

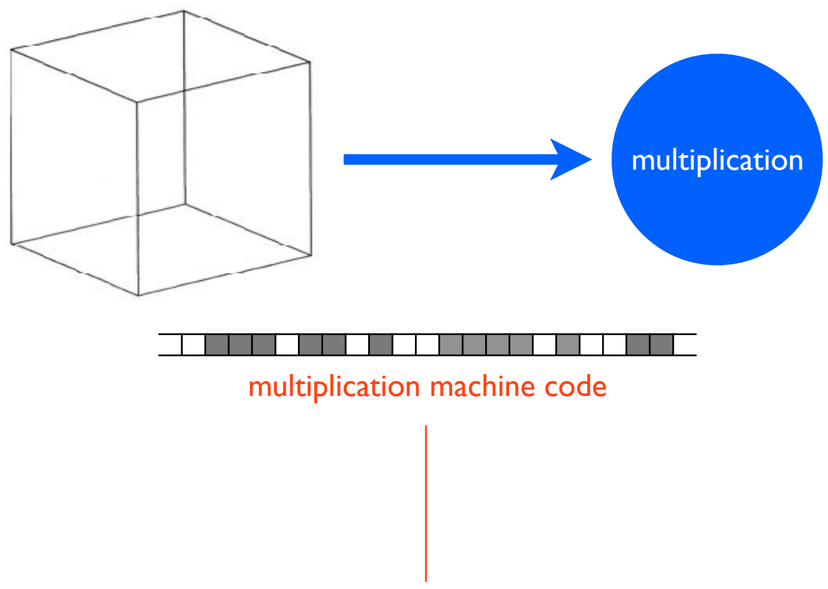
Implicit and Explicit Knowledge

Computes multiplication **without** any representation of a multiplication algorithm.

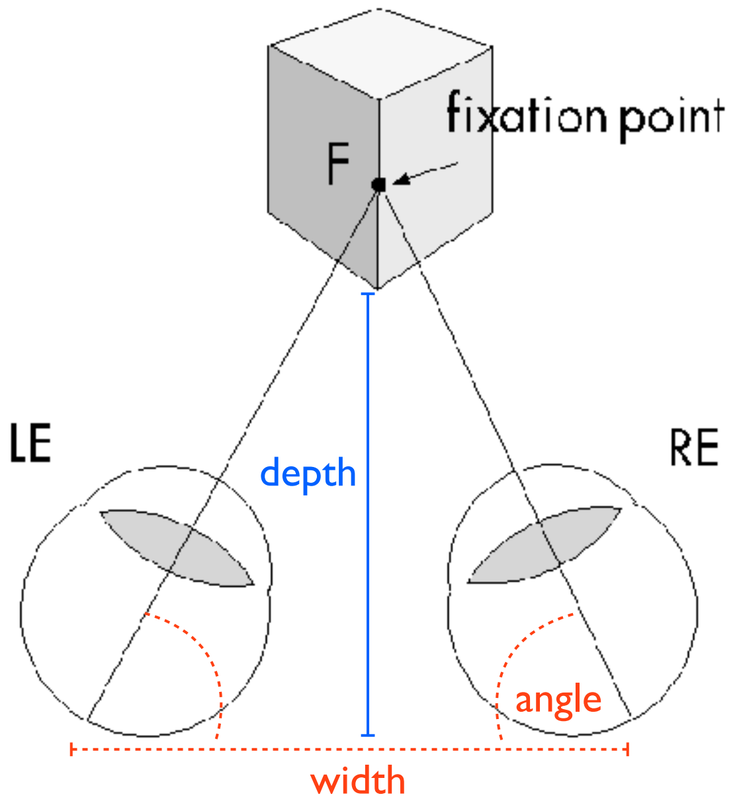


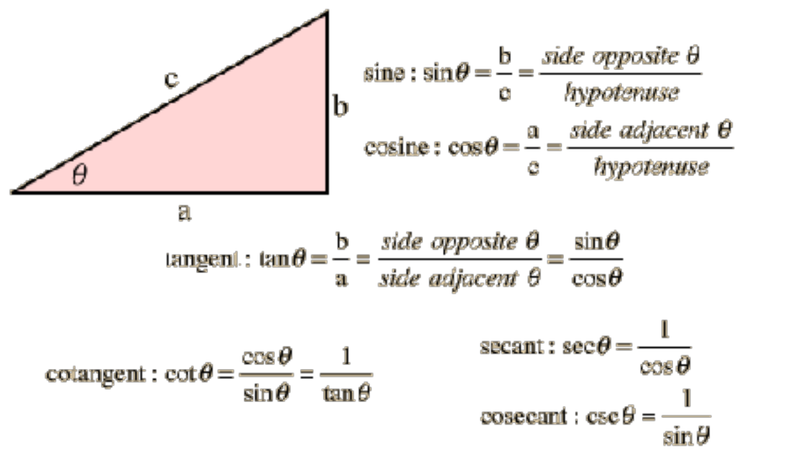
IMPLICIT KNOWLEDGE

Computes multiplication **with** a representation of a multiplication algorithm.



EXPLICIT KNOWLEDGE



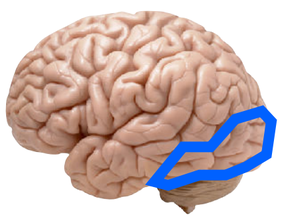




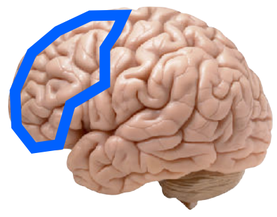
IMPLICIT KNOWLEDGE



EXPLICIT KNOWLEDGE

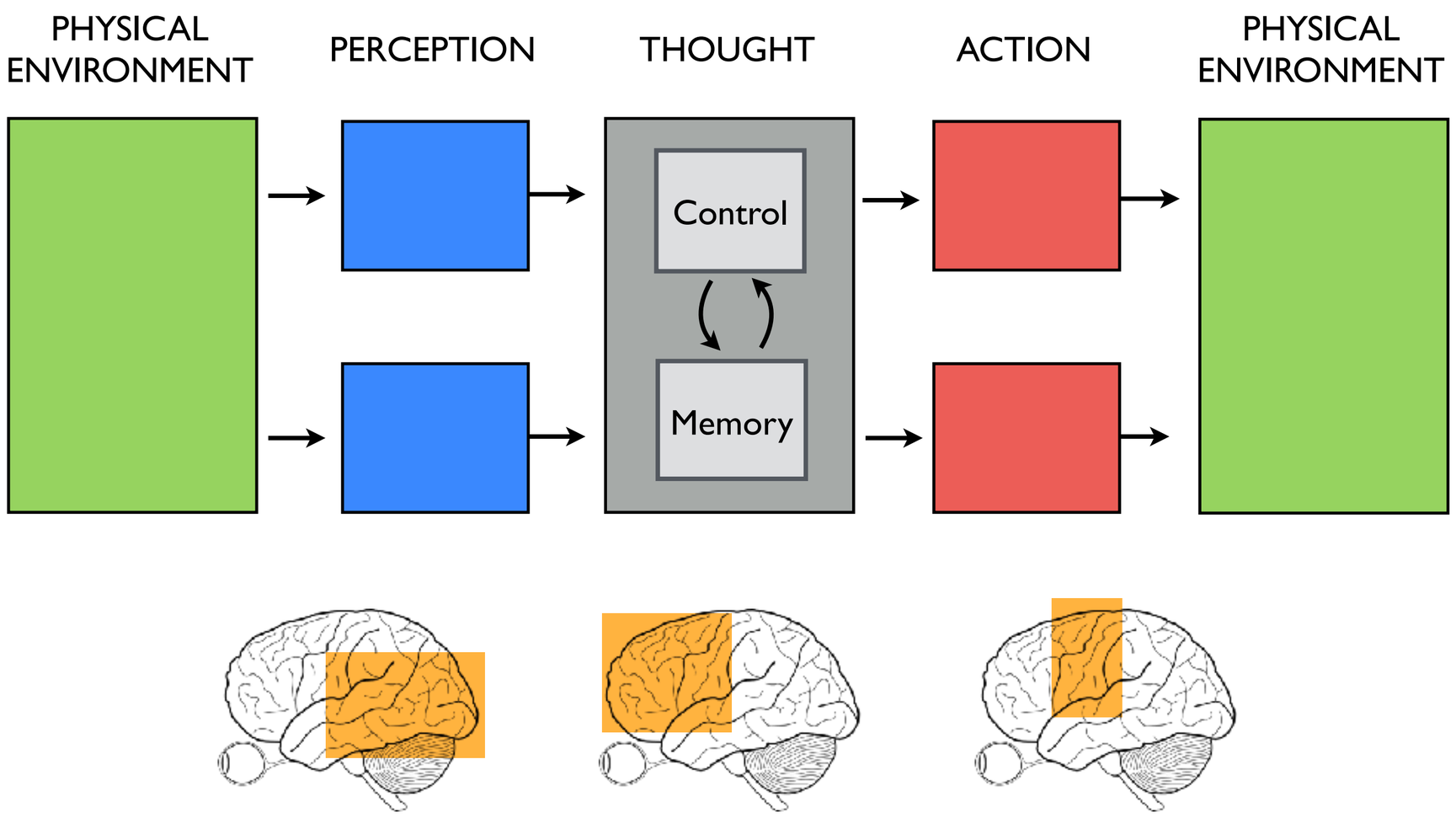


Computes *tangent* **without** an explicit representation of a tangent algorithm.



Computes *tangent* **with** an explicit representation of a tangent algorithm.

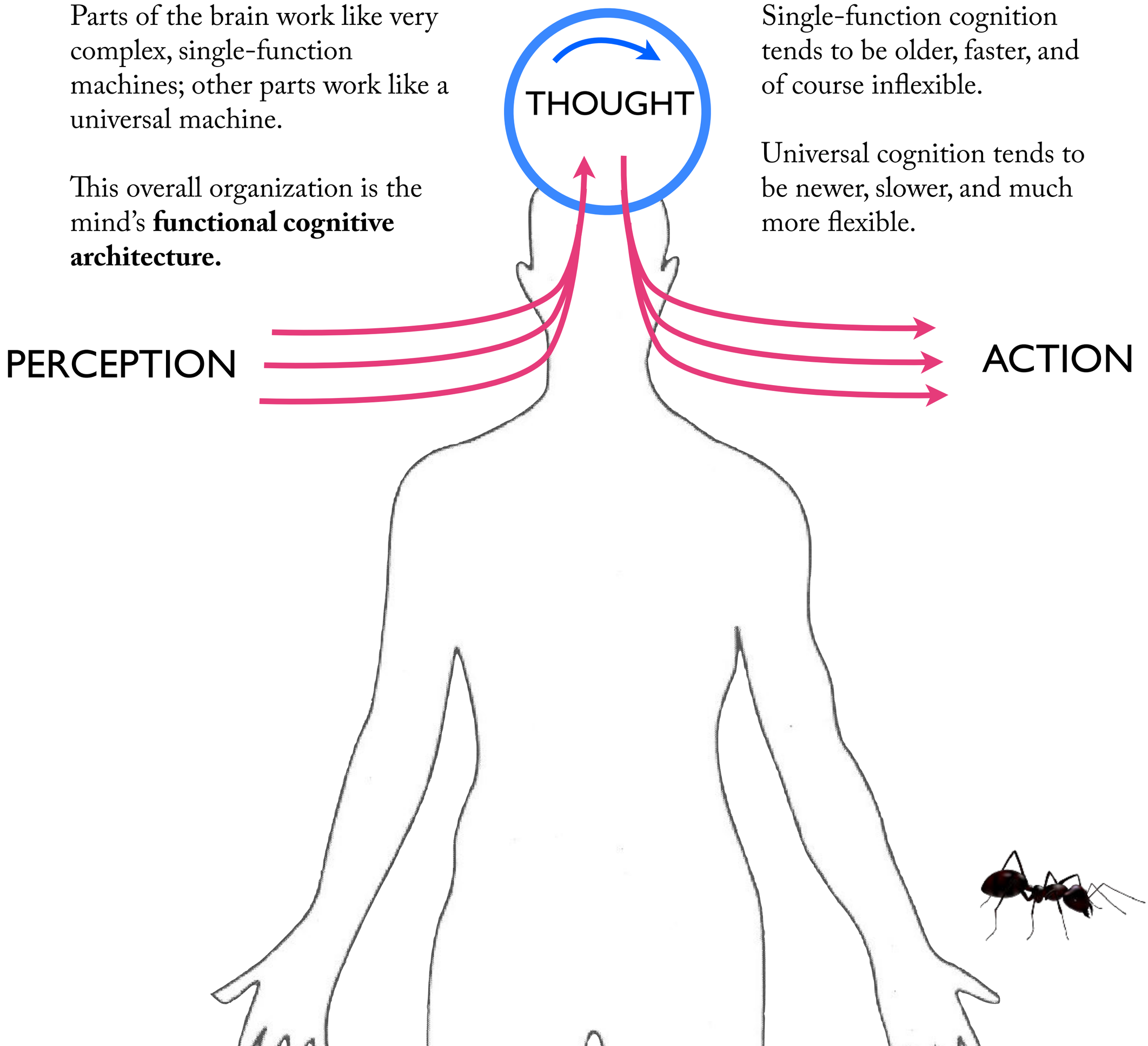
Cognitive Architecture



LOW-LEVEL COGNITION | **HIGH-LEVEL** COGNITION | LOW-LEVEL COGNITION

Parts of the brain work like very complex, single-function machines; other parts work like a universal machine.

This overall organization is the mind's **functional cognitive architecture.**

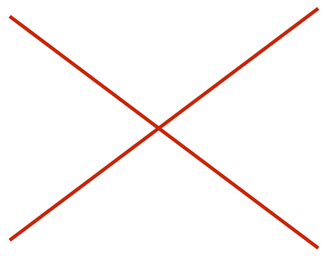


Single-function cognition tends to be older, faster, and of course inflexible.

Universal cognition tends to be newer, slower, and much more flexible.

Implicit and Explicit Knowledge

There are two ways to draw the distinction between "implicit" and "explicit" knowledge.

	Implicit Knowledge Encoded in Architecture (not represented)	**Explicit Knowledge** Encoded in Memory (represented)
Implicit Knowledge Unconscious Knowledge	perceptual rules, motor control rules, knowledge how Turbot: Binary + I	implicit bias, blindsight, ? language rules Turbot: x-y coordinates UTM: Binary + I
Explicit Knowledge Consciously Accessible Knowledge		knowledge that, memories, beliefs, intentions, attitudes, reasoning, mental imagery

Some computer architectures, such as "neural nets" (not necessarily what's in our brains), don't draw a sharp distinction between logic engine and memory. Hence they don't draw a sharp distinction between knowledge encoded in *architecture* and knowledge encoded in *representations*.

